

Azure AKS Infrastructure - Cost Summary

Generated: March 17, 2026
Region: East US
Instance Type: Standard_D4s_v5 (4 vCPU, 16GB RAM)
Currency: USD

Cost Overview

Total Cost Estimates

Configuration	3-Day Cost	Monthly Cost	Cost per Hour
Base (7 nodes)	\$145.73	\$1,457.20	\$2.02/hr
Average (11 nodes)	\$212.77	\$2,127.60	\$2.95/hr
Peak (15 nodes)	\$279.81	\$2,798.00	\$3.88/hr

Recommended Budget for 3-Day Event: \$300 (covers peak with buffer)

Detailed Service Breakdown

1. Compute Resources (AKS Nodes)

VM Type: Standard_D4s_v5 (4 vCPU, 16GB RAM)
Cost per VM: \$0.219/hour | \$158.00/month | \$15.77/3-days

Node Pool	Node Count	3-Day Cost	Monthly Cost	Notes
System Pool	3 nodes	\$47.40	\$474.00	Core services
Frontend Pool	2-8 nodes	\$31.60-\$126.40	\$316.00-\$1,264.00	User-facing apps
Backend Pool	2-7 nodes	\$31.60-\$110.60	\$316.00-\$1,106.00	APIs/services

Scaling Configuration:

- Base Configuration:** 7 nodes total = \$110.60 (3-day) / \$1,106.00 (monthly)
- Average Configuration:** 11 nodes total = \$173.80 (3-day) / \$1,738.00 (monthly)
- Peak Configuration:** 15 nodes total = \$237.00 (3-day) / \$2,370.00 (monthly)

2. AKS Control Plane

Tier	3-Day Cost	Monthly Cost	SLA	Notes
Standard	\$7.30	\$73.00	99.5%	Recommended

3. Storage Costs

OS Disks - Standard SSD (E10, 128 GB per node)

Configuration	Nodes	3-Day Cost	Monthly Cost	Notes
Base	7	\$6.72	\$67.20	\$9.60/disk/month
Average	11	\$10.56	\$105.60	51% cheaper than Premium SSD
Peak	15	\$14.40	\$144.00	500 IOPS per disk

Additional Storage:

- Persistent Volume Claims: \$0.96/100GB (3-day) | \$9.60/100GB (monthly)
- Container Image Storage: Included in ACR (up to 10 GB)

4. Networking

Component	3-Day Cost	Monthly Cost	Notes
Load Balancer (Standard)	\$1.88	\$18.75	Required for AKS
Load Balancer Rules	\$0.80	\$8.00	10 rules
Public IP Address	\$0.37	\$3.65	1 static IP
Data Transfer (outbound)	\$0.00	\$0.00	First 5 GB free
Total Networking	\$3.05	\$30.40	

5. Container Registry (ACR)

Tier	3-Day Cost	Monthly Cost	Storage	Features
Basic	\$1.50	\$15.00	10 GB	Unlimited pulls, Webhooks

Benefits: 95% cost savings vs Premium tier, sufficient for event workloads

6. Monitoring & Logging

Component	3-Day Cost	Monthly Cost	Data Volume	Notes
Log Analytics Workspace	\$16.56	\$165.60	2 GB/day	\$2.76/GB
Container Insights	Included	Included	-	Built-in metrics
Data Retention	\$0.00	\$0.00	31 days	Free included
Total Monitoring	\$16.56	\$165.60	~6 GB (3-day)	

7. Additional Resources (No Cost)

Component	3-Day Cost	Monthly Cost	Notes
Virtual Network	\$0.00	\$0.00	No charge

Component	3-Day Cost	Monthly Cost	Notes
Subnets	\$0.00	\$0.00	No charge
Managed Identity	\$0.00	\$0.00	No charge
Network Security Groups	\$0.00	\$0.00	No charge

Complete Cost Summary by Service

3-Day Event Costs

Service	Base (7 nodes)	Average (11 nodes)	Peak (15 nodes)
AKS Control Plane	\$7.30	\$7.30	\$7.30
Compute (VMs)	\$110.60	\$173.80	\$237.00
Storage (Standard SSD)	\$6.72	\$10.56	\$14.40
Networking	\$3.05	\$3.05	\$3.05
Container Registry	\$1.50	\$1.50	\$1.50
Monitoring & Logs	\$16.56	\$16.56	\$16.56
Additional Resources	\$0.00	\$0.00	\$0.00
TOTAL (3-Day)	\$145.73	\$212.77	\$279.81

Monthly Cost Projection

Service	Base (7 nodes)	Average (11 nodes)	Peak (15 nodes)
AKS Control Plane	\$73.00	\$73.00	\$73.00
Compute (VMs)	\$1,106.00	\$1,738.00	\$2,370.00
Storage (Standard SSD)	\$67.20	\$105.60	\$144.00
Networking	\$30.40	\$30.40	\$30.40
Container Registry	\$15.00	\$15.00	\$15.00
Monitoring & Logs	\$165.60	\$165.60	\$165.60
Additional Resources	\$0.00	\$0.00	\$0.00
TOTAL (Monthly)	\$1,457.20	\$2,127.60	\$2,798.00

Cost Breakdown by Service (Percentage)

3-Day Event - Average Configuration (\$212.77)

Service	Cost	Percentage
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Service	Cost	Percentage
Compute (VMs)	\$173.80	81.7%
Monitoring & Logs	\$16.56	7.8%
Storage	\$10.56	5.0%
AKS Control Plane	\$7.30	3.4%
Networking	\$3.05	1.4%
Container Registry	\$1.50	0.7%

Monthly - Average Configuration (\$2,127.60)

Service	Cost	Percentage
Compute (VMs)	\$1,738.00	81.7%
Monitoring & Logs	\$165.60	7.8%
Storage	\$105.60	5.0%
AKS Control Plane	\$73.00	3.4%
Networking	\$30.40	1.4%
Container Registry	\$15.00	0.7%

Key Insight: Compute resources account for ~82% of total costs. Optimize node count and sizing for maximum savings.

Quick Reference

Hourly Costs

Configuration	Cost per Hour	Daily Cost (24hr)
Base (7 nodes)	\$2.02	\$48.58
Average (11 nodes)	\$2.95	\$70.92
Peak (15 nodes)	\$3.88	\$93.27

Cost Optimization Applied

Optimization	Savings	Impact
Standard SSD vs Premium SSD	51% per disk	\$7.07 (3-day) / \$70.77 (monthly)
Basic ACR vs Premium	95%	\$32.10 (3-day) / \$321.00 (monthly)
Single environment vs Dual	~50%	~\$145 (3-day) / ~\$1,450 (monthly)

Optimization	Savings	Impact
Event-optimized scaling	Variable	Cost scales with actual usage

Budget Recommendations

Duration	Recommended Budget	Configuration Covered
3-Day Event	\$300	Peak + 7% buffer
1 Week	\$550	Average configuration
1 Month	\$2,300	Average configuration

Note: All costs are estimates based on East US region pricing as of March 2026. Actual costs may vary based on usage patterns, data transfer, and Azure pricing changes. Monitor costs using Azure Cost Management for accurate tracking.